lcardoso@larc.usp.br
mywebsite-todo
Lucas
Orcid

Lucas Cupertino Cardoso

Biography

I am a master's student at the Computer and Electrical Engineering Undergraduate Program of the University of São Paulo, where I also obtained my graduation. My main focus is on Applied Cryptography. My master's thesis focuses on using cryptographic primitives (digital signatures and zero-knowledge proofs) to produce secure and traceable compact credentials for distributed systems, especially cloud-based environments. In the future, I will be a doctoral student at the same University, working on Post-Quantum Cryptography. Besides the previously mentioned topics, I am also interested in Theoretical Cryptography, Complexity Analysis, and Computational Number Theory.

Research Interests

- Applied Cryptography.
- Digital Signatures.
- Post-quantum cryptography algorithms.
- Zero-knowledge proofs.
- Cryptanalysis.
- Formal analysis of protocols.
- Automated security proofs (Veriproof, Scyther, Tamarin, etc)

Peer-reviewed Journals

- SchoCo: Schnorr Concatenation for Extensible Tokens. Marco A. Marques, **Lucas C. Cardoso**, Pedro B. Correia, Charles C. Miers, Marco A. Simplicio Junior. preprint. To appear in 2026. Equal contribution in all aspects

Peer-reviewed Conferences

- Next-Generation SPIFFE/SPIRE Identity Management Systems with Post-Quantum Cryptography Algorithms. **Lucas C. Cardoso**, Marco A. Marques, Pedro B. Correia, Henrique Z. Cochak, Charles C. Miers, Marcos A. Simplicio Junior. *The 25th IEEE International Symposium on Cluster, Cloud, and Internet Computing, Tromsø, Norway, 19th–22nd May, 2025*. Equal contribution in all aspects. [1]
- Introducing Two ROS Attack Variants: Breaking One-More Unforgeability of BZ Signature Scheme. Bruno Foschiani, **Lucas C. Cardoso**, Leonardo T. Kimura, Paulo L. Barreto, Marcos A. Simplicio Junior. *The XXV Brazilian Cybersecurity Symposium, Foz do Iguaçú, Brazil, 1st–4th September, 2025*. Equal contribution in all aspects.
- On the Use of ECDSA with Delegated Public Key in Identity-Based Scenarios. **Lucas C. Cardoso**, Marcos A. Simplicio Junior. preprint. To appear in 2026. Equal contributions in all aspects.

Work in Progress

- Compact Credentials with Traceability in Federated Environments. **Lucas C. Cardoso**, Marcos A. Simplicio Junior.

- A Security Classification for Aggregate Signatures. **Lucas C. Cardoso**, Marco A. Marques, Henrique Z. Cochak, Charles C. Miers, Marcos A. Simplicio Junior.
- Providing a Trustful Audit in Instant Messaging Apps using Hash Chain. Andrea E. Komo, **Lucas C. Cardoso**, Leonardo Kimura, Marcos A. Simplicio Junior.

Other Publications

Projects

- Secure Federated Identity Management: Enhancing and Extending the SPIFFE Architecture. Hewlett Packard Enterprise. January 2023 – December 2023.
- Public Security Tests - Superior Electoral Court, Brazil. January 2024 – today.
- Future Elections - Superior Electoral Court, Brazil. January 2024 – today.
- Post-quantum cryptography algorithms. BRADESCO.

List of Co-authors (alphabetically)

References

[1] Lucas Cardoso et al. "Next-Generation SPIFFE/SPIRE Identity Management Systems with Post-Quantum Cryptography Algorithms". In: *The 25th IEEE International Symposium on Cluster, Cloud and Internet Computing*. 2025.